# Martin Špetlík

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## Personal Profile

Applied ML researcher and Ph.D. candidate with experience in scientific machine learning, graph neural networks, and advanced generative modeling for medical imaging. Developed VQ-VAE, VQ-GAN, and diffusion-based architectures to generate complex medical 3D CT scans, and built ML-based surrogates for complex physical simulation processes. Strong background in programming, statistics and numerical methods. Proven ability to identify research problems, survey the literature, design and implement tailored ML or computational solutions. Published researcher with an interdisciplinary mindset and a strong motivation to apply ML to challenging real-world problems.

## Education

**Ph.D. in Applied Sciences in Engineering** — *Liberec, Czech Republic*
Technical University of Liberec — *Sep 2020 – Mar 2026 (Expected)*

- Supervisor: Assoc. Prof. Dr. Jan Březina (jan.brezina@tul.cz).
- Doctoral Research: Pioneered the application of Multilevel Monte Carlo (MLMC) methods for stochastic modeling in fractured rock, an approach made feasible by the integration of machine learning surrogates.

**GeMSS - Generative Modeling Summer School** — *Eindhoven, The Netherlands*
Eindhoven University of Technology — *June 2024*

- Completed intensive 1-week training focused on generative models including VAEs, GANs, diffusion models, or transformer-based architectures.

**Master's Degree in Information Technology** — *Liberec, Czech Republic*
Technical University of Liberec — *July 2020*

- Thesis: *Reconstruction of Distribution from Estimated Generalized Moments*.
- Graduated Cum Laude; recipient of the Dean's Award for Outstanding Master's Thesis.

**Bachelor's Degree in Information Technology** — *Liberec, Czech Republic*
Technical University of Liberec — *June 2018*

- Thesis: *Application of the Multilevel Monte Carlo Method in Hydrogeology*.
- Graduated Cum Laude; recipient of the Dean's Award for Outstanding Bachelor's Thesis.

## Work Experience

**Research Assistant** — *Liberec, Czech Republic*
Technical University of Liberec — *Sep 2023 – Present*

- Architected the machine learning strategy for a generative AI project focused on 3D CT scans of pelvic bones, including model design (VQ-VAE, VQ-GAN, diffusion models), data preprocessing pipelines, and specialized techniques for low-sample, high-dimensional biomedical data.
- Executed large-scale model training and experimentation on HPC systems, optimizing job scheduling, memory usage, and runtime performance.
- Integrated an Unscented Kalman Filter (UKF), using FilterPy, with a nonlinear Richards-equation model for data assimilation, designing the coupling logic, state mappings, and full workflow containerization (Docker) for reproducibility.
- Lead developer and primary maintainer of the open-source `mlmc` Python package (`https://pypi.org/project/mlmc/`).
- Developed and managed the full curriculum, lecture materials, assignments, and exams for the Computer Architecture course; designed seminar content for the Graph Theory course.

**Research Intern** — *Mol, Belgium*
Belgian Nuclear Research Center (SCK CEN) — *Feb 2023 – Jul 2023*

- Mentor: Dr. Eric Laloy (eric.laloy@sckcen.be).
- Designed and implemented a deep-learning surrogate model for numerical homogenization, including dataset construction, neural architecture selection, validation procedures, and deployment for HPC-based workflows.

**Junior Researcher** — *Liberec, Czech Republic*
Technical University of Liberec — *Jan 2020 – Dec 2022*

- Introduced Graph Neural Networks (GNNs) into the research group to accelerate hydrogeological simulations.
- Architected and implemented ML-based meta-models for integration within Multilevel Monte Carlo (MLMC) frameworks to improve simulation efficiency for non-fractured subsurface models.

**Python developer (part-time)** — *Liberec, Czech Republic*
Technical University of Liberec — *Jul 2018 - Dec 2019*

- Developed the initial version of the open-source `mlmc` Python package, establishing its core architecture and feature set.

**Back-End Developer (part-time)** *Liberec, Czech Republic*
Hotel.cz *Jan 2017 - Jun 2018*
- Developed and maintained back-end systems for web applications using PHP, SQL, and Linux servers.

# Key Research Projects

**Generative AI for 3D CT Scans** *Liberec, Czech Republic*
Czech Science Foundation, Grant No. 24-10862S *2024 - Present*
- Led the machine learning strategy and architectural design for generative modeling of 3D CT scans of pelvic bones.
- Developed a specialized pipeline combining VQ-GAN and Denoising Diffusion Models to address challenges of low-sample, high-dimensional medical-imaging data.

**Multilevel Monte Carlo (MLMC) Acceleration via Deep-Learning Surrogates** *Liberec, Czech Republic*
SGS-2025-3560, SGS-2023-3379; EURAD-2 (Grant Agreement ID 101166718) *2023 – Present*
- Principal Investigator of two student grants focused on ML-based acceleration of stochastic simulations; Research Team Member for European and National research grants.
- Designed and deployed convolutional surrogate models to replace computationally expensive numerical homogenization in 2D and 3D fracture–matrix systems, yielding confirmed $100\times$ speedups.

**Data Assimilation in Non-Linear Models** *Liberec, Czech Republic*
Technology Agency of the Czech Republic, SS06010280 *2023 - Present*
- Responsible for data assimilation using Unscented Kalman Filters (UKF) in a nonlinear Richards-equation model.
- Designed experimental setups and integration strategies for UKF-based approaches applied to nonlinear simulation models.

**Open-Source Software Development, UQ, and GNN Integration** *Liberec, Czech Republic*
Technology Agency of the Czech Republic (TH03010227, TK02010118); EURAD (Grant Agreement ID 847593) *2018 – Present*
- Lead contributor to the open-source `mlmc` Python package, responsible for its architecture, development, and integration with external tools.
- Applied Uncertainty Quantification (UQ) methods to groundwater processes and introduced Graph Neural Networks (GNNs) as surrogates for accelerating UQ analyses.
- Ensured software portability, reproducibility, and maintainability for long-term multi-partner scientific projects.

# Skills

**Deep Learning** Multilayer Perceptrons (MLPs), Convolutional Neural Networks (CNNs), Graph Neural Networks (GNNs), Generative AI (VAEs, GANs, Diffusion Models), Vector Quantization (VQ-VAE, VQ-GAN)
**Scientific Computing** Multilevel Monte Carlo (MLMC), Surrogate Modeling, Uncertainty Quantification (UQ), Kalman Filters (including Unscented KF), High-Dimensional Simulation Modeling, High-Performance Computing (HPC; job scheduling, GPU-based training)
**Programming & Tools** Python (PyTorch, TensorFlow, Spektral, NumPy, Pandas, Scikit-learn), MLflow, Git, Linux, Docker, SQL, LaTeX
**Research & Development** Reproducible research, Experimental design, Model development and evaluation, Workflow automation, Technical documentation, Independent problem-solving, Literature-driven method development
**Soft Skills** Team collaboration, Critical thinking, Technical communication, Time management, Teaching and mentoring

# Publications

## Journal Articles

Convolutional Surrogate for 3D Discrete Fracture-Matrix Tensor Upscaling
Martin Špetlík, Jan Březina
*PREPRINT*. 2025, URL: `https://martinspetlik.github.io/assets/pdf/convolutional_surrogate_3D_dfm_tensor_upscaling_preprint.pdf`

Deep learning surrogate for predicting hydraulic conductivity tensors from stochastic discrete fracture-matrix models
Martin Špetlík, Jan Březina, Eric Laloy
*Computational Geosciences*. Springer, 2024, URL: `https://link.springer.com/article/10.1007/s10596-024-10324-8`

Groundwater Contaminant Transport Solved by Monte Carlo Methods Accelerated by Deep Learning Meta-Model
Martin Špetlík, Jan Březina
*Applied Sciences*. MDPI, 2022, URL: `https://www.mdpi.com/2076-3417/12/15/7382`

## Conference Proceedings

Groundwater flow meta-model for multilevel Monte Carlo methods
Martin Špetlík, Jan Březina
*CEUR Workshop Proceedings*, 2021, URL: `https://ceur-ws.org/Vol-2962/paper40.pdf`

## Languages

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|---|---|
| **Czech** | Native |
| **English** | Full Professional Proficiency |
| **German** | Limited Working Proficiency |